

Chapter 28: Special Relativity

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PHY201
Lecture 28



Part I

Einstein's Postulates and Reference Frames

Outline of Part I: Einstein's Postulates and Reference Frames

Introduction

Einstein's Postulates

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Introduction

Einstein's Postulates

Special Relativity: The Big Picture

In 1905, Einstein published his **Special Theory of Relativity**, overturning centuries of intuition about space and time. Two simple postulates force us to abandon absolute time and absolute length.

Why It Matters

- ▶ GPS satellites require relativistic corrections for accuracy
- ▶ Particle accelerators depend on relativistic dynamics daily
- ▶ Nuclear energy and $E = mc^2$ power stars and reactors
- ▶ Cosmic-ray muons reach Earth only because of time dilation

Not “Everything Is Relative”

Relativity identifies what is *absolute*: the **laws of physics** and the **speed of light**.

Different observers disagree on measurements, but they agree on the laws of physics.

Chapter Roadmap

- ▶ Einstein's postulates — the foundation
- ▶ Simultaneity & time dilation — moving clocks run slow
- ▶ Length contraction — moving objects are shorter
- ▶ Relativistic velocity addition — speeds never exceed c
- ▶ Relativistic momentum — $p = \gamma mu$
- ▶ Relativistic energy — $E = \gamma mc^2$ and $E_0 = mc^2$

Organizing Idea

Time dilation, length contraction, and velocity addition are *not* separate phenomena. They all arise from one geometric structure: the **Lorentz transformation** of spacetime.

Outline of Part I: Einstein's Postulates and Reference Frames

Introduction

Einstein's Postulates

Reference Frames

Inertial Reference Frame

A frame in which **Newton's first law holds**: an object at rest stays at rest and one in motion continues at constant velocity unless acted on by a net force.

Frame	Inertial?
Laboratory (approx.)	Yes
Constant-velocity train	Yes
Car turning a corner	No
Accelerating rocket	No
Earth's surface (approx.)	Approximately

Key Principle

There is *no experiment you can perform inside a closed inertial frame* that detects your uniform motion. You cannot distinguish “at rest” from “moving at constant velocity.”

Misconception

A moving frame can still be inertial. Motion itself is not the problem; **changing velocity** (acceleration) breaks the inertial condition.

Galilean Relativity: What Newton Already Knew

Galilean Principle of Relativity (pre-Einstein)

The laws of *mechanics* are the same in all inertial frames.

Velocities add according to: $u = v + u'$

Example: Ball on a Train

- ▶ Train moves at $v = 30 \frac{\text{m}}{\text{s}}$
- ▶ Passenger throws ball at $u' = 20 \frac{\text{m}}{\text{s}}$ forward
- ▶ Ground observer sees: $u = 30 + 20 = 50 \frac{\text{m}}{\text{s}}$

Quick Vote

If you run toward a flashlight beam at v , should you measure light faster than c ?
(Classical answer: yes, $c + v$.)

Newton's Resolution

Newton's laws work perfectly with Galilean addition.

The conflict arises *only* when we ask: does electromagnetism obey the same relativity principle?

Einstein's Answer

Yes — *all* of physics must satisfy the relativity principle, including electromagnetism. This forces a new velocity transformation.

The Crisis: Two Theories That Disagreed

By 1900, physics had two extremely successful theories, but they appeared incompatible:

Newtonian Mechanics

- ▶ Galilean velocity addition: $u = v + u'$
- ▶ Time is absolute (same for all observers)
- ▶ Works for everyday speeds

Maxwell's Electromagnetism

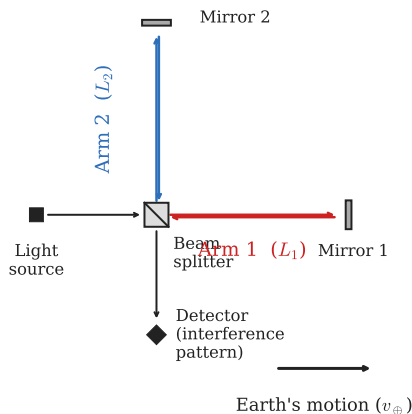
- ▶ Predicts $c = 3.00 \times 10^8 \frac{\text{m}}{\text{s}}$
- ▶ Does *not* specify relative to what
- ▶ Implies c is the same in every frame

The Conflict

If light travels at c in one frame, Galilean addition predicts $c \pm v$ in another, but Maxwell's equations forbid this.

Newton is not **wrong**; his is just the low-speed approximation of a deeper theory.

The Michelson–Morley Experiment (1887)



Result: No fringe shift detected — c is the same in all directions

The Idea

Scientists expected light to behave like sound waves traveling through air. If light traveled through a “luminiferous aether,” Earth’s orbital motion should produce a measurable speed difference between perpendicular arms of the interferometer.

The Result

No difference was found.

The speed of light is the same in all directions, regardless of Earth’s motion through space.

Einstein's Two Postulates (1905)

First Postulate: Principle of Relativity

The laws of physics are the same in **all inertial reference frames**.

No experiment can distinguish one inertial frame from another.

Second Postulate: Constancy of the Speed of Light

The speed of light in a vacuum is the same for all observers, regardless of the motion of the source or the observer:

$$c = 3.00 \times 10^8 \frac{\text{m}}{\text{s}}$$

The Key Inversion

Observers disagree about distance and time so they can agree about c .

Einstein found that abandoning **absolute time and absolute length** was the only consistent choice.

Consequences of the Two Postulates: A Preview

Einstein's two postulates force three kinematic consequences. Each can be derived from the postulates alone — no additional assumptions needed.

Effect	Formula	Physical Meaning
Time dilation	$\Delta t = \gamma \Delta t_0$	Moving clocks run slow
Length contraction	$L = L_0/\gamma$	Moving objects shorter
Velocity addition	$u = (v + u')/(1 + vu'/c^2)$	$u < c$ always

All Three Follow From Postulate 2

If c is constant in all frames, then time and space must be measured differently by different observers. The Lorentz factor γ appears in *all three* formulas — they are not independent effects.

Misconception: What “Same Speed” Means

Common Misconception

Observers don't measure the same speed of light by *correcting* their readings. They each directly measure c with no corrections applied.

What the Second Postulate Requires

- ▶ Observer at rest measures light at c
- ▶ Observer moving toward the source at v also measures c
- ▶ Not $c + v$, not $c - v$ — *exactly* c
- ▶ This forces space and time to be **flexible**

The Consequence

If everyone measures the same c , then **clocks and rulers must disagree** between frames. Time and space are not absolute — they depend on the observer's motion.

Check Your Understanding

A spaceship moves away from Earth at $v = 0.60c$ and turns on a headlight. An Earth observer measures the speed of the emitted light. What do they find?

(A) $0.40c$ (B) $0.60c$ (C) c (D) $1.60c$

Come to lecture to see the answer.

Part II

Time Dilation and Length Contraction

Outline of Part II: Time Dilation and Length Contraction

Simultaneity and Time Dilation

Length Contraction

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Length Contraction

Events and Spacetime

Definition: Event

An **event** is something that happens at a specific *location* and a specific *time* — a point in spacetime (x, t) .

Why Events Matter

Relativity is fundamentally about comparing *measurements of the same event* made by different observers. Before asking “is time dilated?” we must ask: *which events, measured by whom?*

Relativity of Simultaneity

Two events that are **simultaneous** in one inertial frame are *not* necessarily simultaneous in another frame moving relative to the first.

This is not an illusion — it is a fundamental property of spacetime forced by the constancy of c .

Simultaneity: The Train Thought-Experiment

Lightning strikes the front and rear of a moving train **simultaneously** in the embankment observer's frame.

- ▶ Embankment observer: flashes are simultaneous (equidistant midpoint)
- ▶ Train observer: moving *toward* the front flash and *away* from the rear flash
- ▶ Both observers measure the same c
- ▶ Train observer receives the front flash *first*
- ▶ Train observer concludes: flashes were **not simultaneous**

Causal Chain

Same $c \Rightarrow$ different arrival times
 $\Rightarrow$ different conclusions about simultaneity

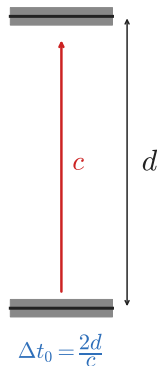
Conclusion

There is no universal “now.”
 Simultaneity is
frame-dependent.

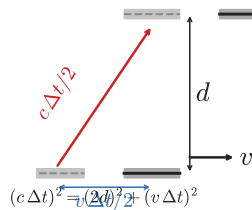
The constancy of c makes this inevitable.

The Light Clock

Clock at rest



Clock moving at v



The Thought Experiment

A “clock” consists of two parallel mirrors. A photon bounces between them. One tick = one round trip.

Prediction

Which path is longer: the vertical path (in the moving clock’s frame) or the diagonal path (seen by the stationary observer)?

If the speed is the same, what must happen to time?

Deriving Time Dilation

Pythagorean Geometry

Let d = mirror separation. One half-tick:

- ▶ Diagonal (stationary observer): $c \Delta t/2$
- ▶ Vertical (proper time): $c \Delta t_0/2 = d$
- ▶ Horizontal: $v \Delta t/2$

Applying the Pythagorean theorem:

$$c^2 \Delta t^2 - v^2 \Delta t^2 = c^2 \Delta t_0^2$$

$$\Delta t^2 (c^2 - v^2) = c^2 \Delta t_0^2$$

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - v^2/c^2}} = \gamma \Delta t_0$$

Result

The stationary observer measures a *longer* time between ticks. The **moving clock runs slow**.

- ▶ Which observer holds the clock?
- ▶ What does Δt_0 represent?

Identifying Proper Time

Decision Tree for Finding Δt_0

- ▶ Are the two events at the same place (in some frame)?
- ▶ If YES: which observer is at that location?
- ▶ That observer measures the proper time Δt_0
- ▶ All other observers measure dilated time $\Delta t = \gamma \Delta t_0$

Apply the Rule

- ▶ Muon created and decays at same location (muon's frame) $\Rightarrow$ muon measures Δt_0
- ▶ Astronaut departs and returns at same location (astronaut's frame) $\Rightarrow$ astronaut measures Δt_0
- ▶ Atomic clock in orbit: events (departure, return) separated in Earth frame $\Rightarrow$ Earth measures $\Delta t > \Delta t_0$

Common Mistake

The rule is about *location*, not speed or whether the observer is “stationary.”

Δt_0 is the **minimum** time interval measured by the observer present at both events.

Time Dilation and Proper Time

Time Dilation

$$\Delta t = \gamma \Delta t_0$$

- ▶ $\Delta t_0 = \mathbf{proper\ time}$: time interval measured by a clock *present at both events* (same location)
- ▶ Δt = dilated time, measured by observer in relative motion
- ▶ $\gamma \geq 1$, so $\Delta t \geq \Delta t_0$ always

Lorentz Factor

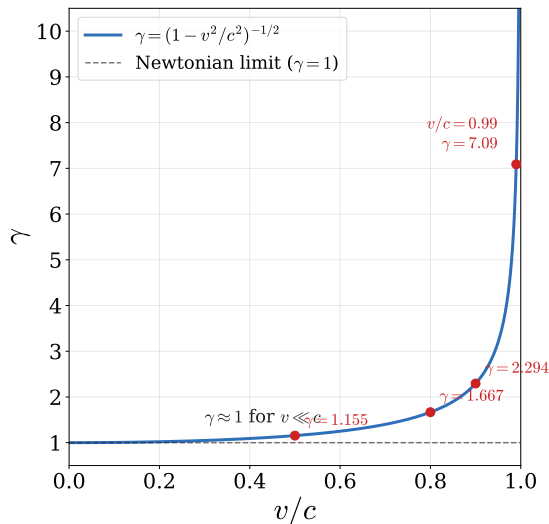
$$\gamma = \left(\sqrt{1 - \frac{v^2}{c^2}} \right)^{-1}$$

- ▶ $v \ll c$: $\gamma \approx 1$ (Newtonian limit)
- ▶ $v \rightarrow c$: $\gamma \rightarrow \infty$

Proper Time

The *shortest* time between two events is always the proper time — measured by the clock present at both events.

The Lorentz Factor γ



Selected Values

v/c	γ
0.10	1.005
0.50	1.155
0.80	1.667
0.90	2.294
0.99	7.089
0.999	22.37

Newtonian Limit

At everyday speeds ($v \ll c$), $\gamma \approx 1$ and relativistic effects are unmeasurable. Classical mechanics is the low- v limiting case.

Time Dilation: Everyday vs. Relativistic Speeds

How large is time dilation at various speeds?

Object / Scenario	v	γ	Extra time / hr
Walking (1.4 m/s)	$4.7 \times 10^{-9}c$	≈ 1	$\sim 10^{-17}$ s
Commercial jet (250 m/s)	$8.3 \times 10^{-7}c$	≈ 1	$\sim 10^{-12}$ s
ISS astronaut (7700 m/s)	$2.6 \times 10^{-5}c$	≈ 1	28 μ s/day
GPS satellite (3900 m/s)	$1.3 \times 10^{-5}c$	≈ 1	7 μ s/day (SR)
Muon from cosmic ray	$0.999c$	22.4	$\times 22.4$
LHC proton	$0.9999991c$	7460	$\times 7460$

Key Takeaway

At everyday speeds, $\gamma \approx 1$ — relativistic effects are completely negligible. Newtonian mechanics is an excellent approximation for $v \lesssim 0.1c$.

Example: Cosmic Muon Decay

Muon Survival — Time Dilation

Muons created at $h = 10.0$ km altitude travel at $v = 0.999c$. Proper half-life: $\Delta t_0 = 2.20 \mu\text{s}$.
 (a) What half-life do ground observers measure? (b) How far do muons travel in one ground-frame half-life?

$$\begin{aligned}\gamma &= \frac{1}{\sqrt{1 - 0.999^2}} \\ &= \frac{1}{\sqrt{0.001999}} = 22.4\end{aligned}$$

$$\begin{aligned}\text{(a)} \quad \Delta t &= \gamma \Delta t_0 \\ &= 22.4 \times 2.20 \mu\text{s}\end{aligned}$$

$$\Delta t = 49.3 \mu\text{s}$$

$$\begin{aligned}\text{(b)} \quad d &= v \Delta t \\ &= 0.999c \times 49.3 \mu\text{s}\end{aligned}$$

$$d = 14.8 \text{ km} > h$$

Without time dilation, muons travel only
 $0.999c \times 2.20 \mu\text{s}$
 $= 659 \text{ m}$ — far short of 10 km.
 Time dilation explains why muons reach Earth's surface.

► In the muon's frame, is it time dilation or length contraction that explains this?

Example: Astronaut Time Dilation

Moving Clocks

An astronaut travels at $v = 0.990c$ relative to Earth. The astronaut's clock measures $\Delta t_0 = 1.00$ yr. How much time elapses on Earth?

$$\gamma = \frac{1}{\sqrt{1 - 0.990^2}} = \frac{1}{\sqrt{0.0199}}$$

$$\gamma = 7.09$$

$$\Delta t = \gamma \Delta t_0$$

$$\Delta t = 7.09 \times 1.00 \text{ yr}$$

$$\Delta t = 7.09 \text{ yr}$$

For every year the astronaut experiences, over 7 years pass on Earth.

- If the astronaut returns to Earth, who is older?
- Why is this not a paradox?

The Twin Paradox

Twin A stays on Earth. Twin B travels at high speed to a distant star and returns. Who is older when they reunite?

- ▶ Naively: each twin sees the other moving — aren't the situations symmetric?
- ▶ **The asymmetry:** Twin B must *accelerate* to turn around. The two situations are physically distinguishable.
- ▶ During the turnaround, Twin B is in a non-inertial frame.
- ▶ **Result:** Twin B (the traveler) is younger.

Not a Paradox

The apparent symmetry breaks because only Twin B accelerates. The two frames are *not* equivalent.

Confirmed Experimentally

Atomic clocks flown around the world fall behind ground-based clocks — exactly as predicted.

What Makes Time Dilation Unavoidable

Time dilation is not a choice or a convention. It emerges inevitably from the two postulates.

The Logic Chain

- ▶ All inertial observers measure the same c (Postulate 2)
- ▶ A moving observer sees a longer light-clock path
- ▶ But c is the same, so the time *must* be longer
- ▶ Therefore: $\Delta t = \gamma \Delta t_0$ is inevitable

This Is Not Opinion

Given the two postulates, time dilation *must* occur. It is not a matter of perspective or interpretation.

Why γ Matters

The factor $\gamma = 1/\sqrt{1 - v^2/c^2}$ is not arbitrarily chosen. It emerges directly from the geometry of spacetime.

All of special relativity depends on this one formula.

Experimental Confirmation of Time Dilation

Hafele–Keating Experiment (1971)

- ▶ Four cesium atomic clocks flown around the world on commercial jets
- ▶ Compared with ground-based clocks
- ▶ SR predicts: flying clocks lose time
- ▶ GR predicts: flying clocks gain time (altitude)
- ▶ Net: -59 ± 10 ns
- ▶ Measured: -59 ± 10 ns ✓

Muon Lifetime

Muon lifetime in accelerators matches the $\gamma \tau_0$ prediction to better than 0.1%.

GPS Clocks

GPS satellites must apply both SR and GR corrections. Without them, position errors would exceed $10 \frac{\text{km}}{\text{d}}$.

Time dilation is not a theoretical curiosity; it is measured daily in operational systems.

Relativistic Doppler Effect

Relativistic Doppler Shift

For a source moving directly away from the observer at speed u :

$$\lambda_{\text{obs}} = \lambda_s \sqrt{\frac{1 + u/c}{1 - u/c}}$$

- ▶ Moving away: $\lambda_{\text{obs}} > \lambda_s$ (redshift)
- ▶ Moving toward: replace u with $-u$ (blueshift)
- ▶ Includes both the classical Doppler effect *and* time dilation

Application: Cosmology

Distant galaxies show redshifted spectral lines. The relativistic Doppler formula allows astronomers to measure recession speeds, confirming the expansion of the universe.

Transverse Doppler

A source moving *perpendicular* to the line of sight still shows a redshift — purely due to time dilation. There is no classical analogue.

Example: Galactic Recession from Doppler

Hubble Redshift

A galaxy emits hydrogen light at $\lambda_s = 121.6 \text{ nm}$ (the Lyman- α line). Earth observers measure $\lambda_{\text{obs}} = 243.2 \text{ nm}$. How fast is the galaxy receding?

$$\frac{\lambda_{\text{obs}}}{\lambda_s} = \sqrt{\frac{1 + u/c}{1 - u/c}}$$

$$\frac{243.2}{121.6} = \sqrt{\frac{1 + u/c}{1 - u/c}} = 2.00$$

$$4.00 = \frac{1 + u/c}{1 - u/c}$$

$$4(1 - u/c) = 1 + u/c$$

$$3 = 5u/c$$

$$u = 0.600c$$

The galaxy recedes at 60% of the speed of light. This is a **cosmological redshift** — the wavelength doubled means the galaxy is at a cosmological distance.

► For $z = \lambda_{\text{obs}}/\lambda_s - 1 = 1$, what is u/c ?

Check Your Understanding

A spaceship clock ($v = 0.800c$) ticks off $\Delta t_0 = 5.00$ s of proper time. How much time passes on Earth?

($\gamma = 1.667$ at $v = 0.800c$)

Come to lecture to see the answer.

Outline of Part II: Time Dilation and Length Contraction

Simultaneity and Time Dilation

Length Contraction

Length Contraction: The Concept

If a moving clock runs slow, consistency of speed = distance/time across frames requires that *moving objects are also shorter*.

Length Contraction

$$L = \frac{L_0}{\gamma} = L_0 \sqrt{1 - \frac{v^2}{c^2}}$$

- ▶ L_0 = **proper length**: length in rest frame
- ▶ L = contracted length (observer in relative motion)
- ▶ $L \leq L_0$ always
- ▶ Contraction only along direction of motion

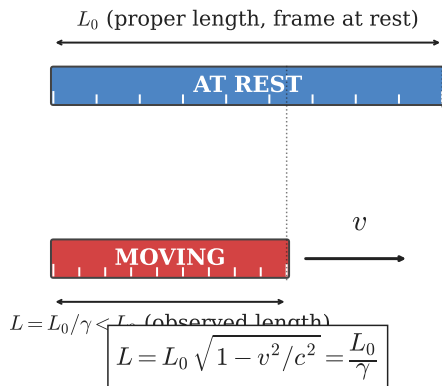
Direction Matters

Perpendicular dimensions are unchanged. Only the component along $\vec{v}$ contracts.

Two Sides of One Coin

From the muon's frame: the atmosphere is length-contracted to ≈ 450 m.

Proper Length and the Contracted Ruler



Proper Length L_0

The length of an object measured in the frame where it is **at rest**. This is always the *longest* measured length.

Simultaneity Connection

To measure a moving rod's length, you must record both endpoints at the *same time*. But simultaneity is frame-dependent — different observers disagree about “same time,” producing different length measurements.

Two Observers, One Muon: Same Physics

The survival of cosmic muons can be explained in two equivalent ways, depending on which frame you adopt.

Frame	Explanation	Numbers
Earth frame	Muon half-life is dilated: $\Delta t = \gamma \Delta t_0$	$22.4 \times 2.20 \mu\text{s} = 49.3 \mu\text{s}$
Muon's frame	Atmosphere is length-contracted: $L = L_0/\gamma$	$10 \text{ km}/22.4 = 447 \text{ m}$

Consistency

Both frames predict the muon survives. Neither is “more correct.” The physics is *frame-independent*; only the *description* changes.

The Deeper Lesson

Time dilation and length contraction are two sides of the same geometric effect in spacetime. γ appears in both because they share the same underlying transformation.

Length Contraction: Misconception Alert

Misconception: Physical Compression

Objects do not shrink because forces squeeze them. There is no mechanical compression, no stress, no deformation.

Correct Interpretation

Length contraction arises because different observers disagree about *which measurements of the endpoints are simultaneous*.

- ▶ Moving observer records endpoints “at the same time” in their frame
- ▶ Stationary observer says those recordings were *not* simultaneous
- ▶ The disagreement in simultaneity directly produces the length difference

Geometric Analogy

A tilted rod appears shorter when projected onto the x -axis. Length contraction is a *geometric effect in spacetime*, not a mechanical one.

Example: Spacecraft Length Contraction

Moving Spacecraft

A spacecraft is $L_0 = 100$ m long at rest. It travels at $v = 0.800c$. What length does an Earth observer measure?

$$\gamma = \frac{1}{\sqrt{1 - 0.800^2}}$$

$$= \frac{1}{\sqrt{1 - 0.640}} = \frac{1}{\sqrt{0.360}}$$

$$\gamma = \frac{1}{0.600} = 1.667$$

$$L = \frac{L_0}{\gamma} = \frac{100 \text{ m}}{1.667}$$

$$\boxed{L = 60.0 \text{ m}}$$

Earth observers measure the craft to be only 60% of its rest length. The crew aboard measure nothing unusual — they are at rest with respect to the craft.

► What does the spacecraft crew measure for Earth's length in the direction of travel?

Check Your Understanding

A space station has proper length $L_0 = 1.00$ km and moves at $v = 0.500c$ relative to Earth. What length do Earth observers measure?

$$(\gamma = 1.155 \text{ at } v = 0.500c)$$

Come to lecture to see the answer.

Example: Length Contraction in Space Travel

Interstellar Journey

A spacecraft travels to Proxima Centauri at $v = 0.950c$. The distance is 4.24 light-years in Earth's rest frame. What do crew members measure for the distance?

$$\gamma = \frac{1}{\sqrt{1 - 0.950^2}} = 3.203$$

$$L = \frac{L_0}{\gamma} \\ = \frac{4.24 \text{ light-years}}{3.203}$$

$$L = 1.32 \text{ light-years}$$

Due to length contraction, the crew measures the distance as only 1.32 light-years along their direction of travel.

This makes interstellar travel *appear* shorter to the travelers, though clock time still elapses.

► How much time does the journey take in the crew's frame? (At $v = 0.950c$)

Classical vs. Relativistic Limits

Newtonian Limit: $v \ll c$

When velocities are small compared to c :

- ▶ $\gamma \approx 1$
- ▶ $\Delta t \approx \Delta t_0$ (no dilation)
- ▶ $L \approx L_0$ (no contraction)
- ▶ $p \approx mu$ (classical momentum)
- ▶ $E \approx mc^2 + \frac{1}{2}mu^2$ (rest energy + KE)

Relativistic Regime: v comparable to c

- ▶ γ becomes large ($\gamma \rightarrow \infty$ as $v \rightarrow c$)
- ▶ Dramatic time dilation
- ▶ Dramatic length contraction
- ▶ Momentum diverges as $p \rightarrow \infty$
- ▶ Total energy dominates:
 $E \approx \gamma mc^2$

Key Insight

Newton's mechanics is the low-speed approximation to relativity. The two theories agree when $v \ll c$, but diverge dramatically for speeds approaching the speed of light.

Part III

Relativistic Velocity Addition

Outline of Part III: Relativistic Velocity Addition

Relativistic Addition of Velocities

Outline of Part III: Relativistic Velocity Addition

Relativistic Addition of Velocities

Why Galilean Velocity Addition Fails

Galilean addition: $u = v + u'$.

Problem: a spaceship at $v = 0.8c$ firing a laser forward gives $u = 0.8c + c = 1.8c$ — violating the second postulate.

Relativistic Velocity Addition

$$u = \frac{v + u'}{1 + \frac{vu'}{c^2}}$$

- ▶ v = velocity of frame S' relative to S
- ▶ u' = velocity of object in frame S'
- ▶ u = velocity of object observed in frame S

Limit Check: Light

Set $u' = c$ (light in S'):

$$u = \frac{v + c}{1 + v/c} = \frac{c(v + c)}{c + v} = c \checkmark$$

Light always travels at c in every frame.

Newtonian Limit

For $v, u' \ll c$, the denominator $\rightarrow 1$ and $u \rightarrow v + u'$. $\checkmark$

Example: Relativistic Velocity Addition

Two Spaceships

Spaceship A moves at $v = 0.750c$ relative to Earth. It fires a probe *forward* at $u' = 0.500c$ relative to the ship. What speed does Earth observe for the probe?

$$\begin{aligned}
 u &= \frac{v + u'}{1 + \frac{v u'}{c^2}} \\
 &= \frac{0.750c + 0.500c}{1 + \frac{(0.750c)(0.500c)}{c^2}} \\
 &= \frac{1.250c}{1 + 0.375} \\
 &= \frac{1.250c}{1.375}
 \end{aligned}$$

$$u = 0.909c$$

Galilean addition predicts $1.250c > c$.
 Relativistic addition gives $0.909c < c$.
 No combination of sub- c speeds ever reaches c .

► What if the probe is fired *backward* at $u' = -0.500c$?

Speed of Light: The Universal Limit

No massive object can reach or exceed the speed of light.

- ▶ As $v \rightarrow c$: Lorentz factor $\gamma \rightarrow \infty$
- ▶ Momentum $p = \gamma mu \rightarrow \infty$
- ▶ Energy $E = \gamma mc^2 \rightarrow \infty$
- ▶ *Infinite* energy would be needed to reach c
- ▶ No combination of sub- c velocities can exceed c

What CAN travel at c ?

Only **massless** particles: photons and gravitational waves.

For massless particles: $E = pc$, and they travel at exactly c in every frame.

Causality

Faster-than- c travel would allow sending signals backward in time, violating causality — the reason effect cannot precede cause.

Check Your Understanding

Spaceship B moves at $v = 0.600c$ relative to Earth. It fires a projectile *backward* at $u' = -0.400c$ relative to the ship. What speed does Earth observe?

Come to lecture to see the answer.

Why Velocity Addition Must Change: Summary

Relativistic velocity addition solves the fundamental problem: Galilean addition violates the constancy of c .

The Problem

- ▶ Galilean: $u = v + u'$
- ▶ Two $0.9c$ speeds $\Rightarrow u = 1.8c > c$
[WRONG]
- ▶ Violates the second postulate

The Solution

$$u = \frac{v + u'}{1 + \frac{vu'}{c^2}}$$

Two $0.9c$ speeds $\Rightarrow u \approx 0.994c < c$ [OK]

Why It Works

The denominator $(1 + vu'/c^2) > 1$ always reduces the sum below c .
At low speeds: denominator ≈ 1 and we recover Galilean addition.

Insight

Velocity addition is not fundamental; it is a *consequence* of the Lorentz transformation of spacetime coordinates.

Lorentz Transformations: A Preview

What Are Lorentz Transformations?

The **Lorentz transformation** is the rule that converts one observer's measurements (x, t) into another observer's measurements (x', t') in a moving frame.

The Transformations

For a frame S' moving at velocity v relative to S :

$$x' = \gamma(x - vt)$$

$$t' = \gamma\left(t - \frac{vx}{c^2}\right)$$

What This Means

Space and time are *mixed together* in the transformation. They are not separate; they form a unified **spacetime**.

How It Leads to Effects

Time dilation, length contraction, and relativistic velocity addition all follow directly from these transformations.

They are not separate phenomena — they are different aspects of the same geometric structure.

Relativity and Causality

Why is c a universal speed limit? Because faster-than-light signals would violate causality itself.

The Causality Argument

- ▶ Event A: explosion at location x_A , time t_A
- ▶ Event B: light from explosion reaches location x_B
- ▶ If light traveled faster than c , different observers would disagree on which event happened first
- ▶ One observer would see effect before cause — impossible

Deep Principle

The speed of light limit is not a property of light alone. It is a property of the spacetime structure itself.

Causality, that effects follow causes, is woven into the geometry of spacetime.

All Inertial Frames Are Equivalent

The Principle of Relativity

No inertial reference frame is privileged. The laws of physics are the same in all inertial frames.

- ▶ Frame A (Earth) is inertial
- ▶ Frame B (spaceship at $0.99c$) is also inertial
- ▶ Both observers measure the same c
- ▶ Both can perform the same experiments
- ▶ Neither can claim to be “at rest” absolutely
- ▶ Each sees the other’s clock run slow

Mutual Time Dilation

If observer A sees observer B’s clock running slow by factor γ , then observer B sees observer A’s clock running slow by the *same* factor γ .

This is not a contradiction — it reflects the relativity of simultaneity.

Part IV

Relativistic Dynamics

Outline of Part IV: Relativistic Dynamics

Relativistic Momentum

Relativistic Energy

Outline of Part IV: Relativistic Dynamics

Relativistic Momentum

Relativistic Energy

Why Classical Momentum Must Be Revised

Classical momentum $p = mu$ is not conserved if we apply the relativistic velocity transformation. Conservation requires a new definition.

Relativistic Momentum

$$p = \gamma mu$$

- ▶ m = invariant (rest) mass — unchanged by speed
- ▶ u = velocity of the particle
- ▶ $\gamma = \left(\sqrt{1 - u^2/c^2}\right)^{-1}$
- ▶ As $u \rightarrow c$: $p \rightarrow \infty$

Conservation Restored

With $p = \gamma mu$, momentum is conserved in all inertial frames. This is the *only* form consistent with both postulates.

Invariant Mass

Mass is invariant — it is the same in all frames. The Lorentz factor γ is what increases with speed, multiplying the velocity.

The Lorentz Factor: A Geometric Quantity

$\gamma = 1/\sqrt{1 - v^2/c^2}$ is not an ad-hoc correction factor. It is a geometric property of spacetime.

Where γ Appears

- ▶ Time dilation: $\Delta t = \gamma \Delta t_0$
- ▶ Length contraction: $L = L_0/\gamma$
- ▶ Momentum: $p = \gamma m u$
- ▶ Energy: $E = \gamma m c^2$
- ▶ Velocity addition denominator: $1 + v u'/c^2$

All contain γ or closely related factors.

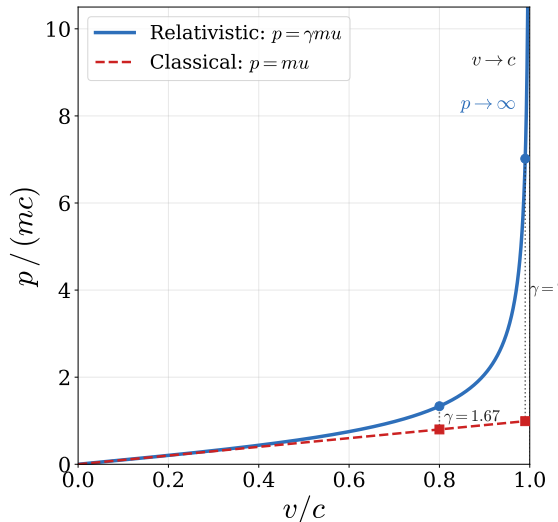
No Accident

The appearance of the same γ in all formulas is not coincidence. It reflects a single underlying geometric truth about spacetime.

Universal

Every relativistic effect comes from the same spacetime geometry. This is why special relativity is so powerful and unified.

Relativistic Momentum vs. Speed



Key Features

- ▶ $v \ll c$: relativistic $\approx$ classical
- ▶ $v \rightarrow c$: $p \rightarrow \infty$ (never reach c)
- ▶ Classical severely underestimates p at high speed

Particle Accelerators

LHC protons travel at $v = 0.9999991c$. Their momentum is ~ 7000 times the classical value. Relativistic mechanics is *essential* for accelerator design.

Example: Relativistic Momentum

Proton Momentum

A proton ($m_p = 1.673 \times 10^{-27}$ kg) moves at $v = 0.800c$. Find its relativistic momentum.

$$\begin{aligned}\gamma &= \frac{1}{\sqrt{1 - 0.800^2}} \\ &= \frac{1}{0.600} = 1.667\end{aligned}$$

$$\begin{aligned}p &= \gamma mu \\ &= 1.667 \times 1.673 \times 10^{-27} \text{ kg} \\ &\quad \times 0.800c \\ &= 1.667 \times 1.673 \times 10^{-27} \\ &\quad \times 2.40 \times 10^8 \frac{\text{m}}{\text{s}}\end{aligned}$$

$$p = 6.69 \times 10^{-19} \text{ kg} \cdot \text{m/s}$$

Classical:

$$p_c = mu = 4.02 \times 10^{-19} \text{ kg} \cdot \text{m/s}$$

Relativistic momentum is $\gamma = 1.667$ times larger than the classical value.

► At $v = 0.990c$ ($\gamma = 7.09$), by what factor does p exceed the classical prediction?

Check Your Understanding

An electron ($m_e = 9.11 \times 10^{-31} \text{ kg}$) moves at $v = 0.900c$. Using $\gamma = 2.294$, find its relativistic momentum.

Come to lecture to see the answer.

Why Nothing Can Reach the Speed of Light

The speed of light is not just a speed limit — it is a geometric limit of spacetime itself.

The Momentum Argument

To reach c , an object with momentum $p = \gamma mu$ would need

$$\gamma \rightarrow \infty \quad \Rightarrow \quad p \rightarrow \infty$$

Infinite momentum requires infinite force or infinite energy.

The Energy Argument

Accelerating a particle requires work: $F = \frac{dp}{dt}$.

To reach c requires:

$$E = \gamma mc^2 \rightarrow \infty$$

Infinite energy is needed but unavailable.

The Lorentz Factor

γ diverges as $v \rightarrow c$. The curve $\gamma(v)$ is not merely steep — it is mathematically *asymptotic*.

Massless Particles

Only objects with $m = 0$ (photons) can travel at c , and they *must* travel at c .

Outline of Part IV: Relativistic Dynamics

Relativistic Momentum

Relativistic Energy

Invariant Mass: What's Truly Conserved

In a collision or reaction, different observers measure different energies and momenta. But they all agree on the *invariant mass*.

For a Single Particle

$$m^2 c^4 = E^2 - (pc)^2$$

- ▶ Different frames measure different E and p
- ▶ But m (the rest mass) is the same in all frames
- ▶ This is the *invariant*

In Collisions

Energy and momentum separately conserved. Total rest mass can change if energy is released (mass-to-energy conversion).

The invariant mass m is what remains constant through all Lorentz transformations.

What Energy Really Means in Relativity

In classical physics: $E = \frac{1}{2}mu^2$ is energy available for work. In relativity: energy and mass are interconvertible.

Three Energy Concepts

- ▶ **Rest energy** $E_0 = mc^2$: energy of a stationary particle
- ▶ **Kinetic energy** $KE = (\gamma - 1)mc^2$: energy due to motion
- ▶ **Total energy** $E = \gamma mc^2$: sum of the two

Mass-Energy Equivalence

A particle with mass m at rest contains energy $E_0 = mc^2$. In nuclear reactions, a tiny fraction of this mass converts to kinetic energy.

Energy Conservation

Total energy is conserved in all reactions and collisions. Energy can be released (as in nuclear fission) or absorbed (as in forming a nucleus), but the total is constant.

Why c^2 Is Huge

$c = 3 \times 10^8 \frac{\text{m}}{\text{s}}$, so c^2 is enormous. Even a tiny mass contains tremendous energy.

Relativistic Kinetic Energy

Relativistic KE

The work done to accelerate a particle from rest to speed u :

$$KE = (\gamma - 1)mc^2$$

- ▶ $v \ll c$: $\gamma - 1 \approx \frac{1}{2}v^2/c^2$, so $KE \approx \frac{1}{2}mv^2$ ✓
- ▶ $v \rightarrow c$: $KE \rightarrow \infty$ — never reach c

Net Work

$$W_{\text{net}} = E - E_0 = KE$$

Why $(\gamma - 1)mc^2$?

Total energy is $E = \gamma mc^2$.

Energy at rest is $E_0 = mc^2$.

Kinetic energy is the difference:

$$KE = E - E_0 = (\gamma - 1)mc^2$$

Rest Energy: $E_0 = mc^2$

The Rest Energy

Even a stationary particle possesses energy:

$$E_0 = mc^2$$

- ▶ Proton at rest: $E_0 = 938 \text{ MeV}$
- ▶ Electron at rest: $E_0 = 0.511 \text{ MeV}$
- ▶ $c^2 = 9 \times 10^{16} \frac{\text{J}}{\text{kg}}$ — mass contains *enormous* energy
- ▶ Nuclear reactions release a tiny fraction ($\sim 0.1\%$) of rest energy — enough to power the Sun

Hot vs. Cold Coffee

A hot cup of coffee has slightly *more mass* than a cold one.

Thermal energy increases rest energy.

The mass difference is far too small to measure, but it is physically real.

Total Relativistic Energy

Total Energy

$$E = \gamma mc^2$$

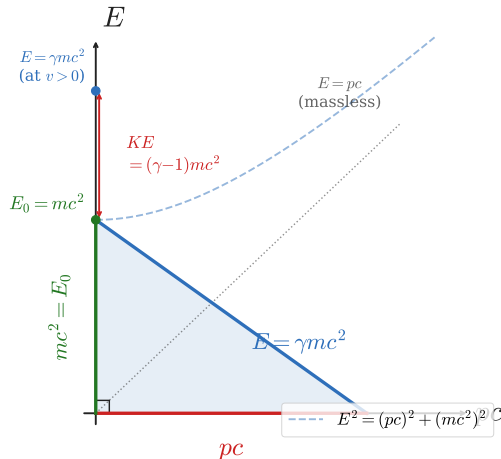
Decomposition: $E = E_0 + KE$

$$\gamma mc^2 = mc^2 + (\gamma - 1)mc^2$$

Energy–Momentum Relation

$$E^2 = (pc)^2 + (mc^2)^2$$

For $m = 0$: $E = pc$.



The Energy Hierarchy

The most fundamental relation is $E^2 = (pc)^2 + (mc^2)^2$, not $E = mc^2$. The famous formula $E_0 = mc^2$ is the *rest-energy special case*.

Quantity	Formula	When
Rest energy	$E_0 = mc^2$	$v = 0$
Kinetic energy	$KE = (\gamma - 1)mc^2$	$v > 0$, massive
Total energy	$E = \gamma mc^2$	any speed
Energy-momentum	$E^2 = (pc)^2 + (mc^2)^2$	any particle
Massless particle	$E = pc$	$m = 0$

Most Important Takeaway

$E = mc^2$ is *not* the whole theory. It is only the rest energy. Total energy $E = \gamma mc^2$ and $KE = (\gamma - 1)mc^2$ govern dynamics.

Example: Electron Kinetic Energy

Relativistic KE

An electron ($m_e = 9.11 \times 10^{-31}$ kg) travels at $v = 0.990c$. Find (a) its kinetic energy and (b) its rest energy.

$$\gamma = \frac{1}{\sqrt{1 - 0.990^2}} = 7.09$$

$$\begin{aligned} \text{(a)} \quad KE &= (\gamma - 1)mc^2 \\ &= 6.09 \times 9.11 \times 10^{-31} \\ &\quad \times (3.00 \times 10^8)^2 \end{aligned}$$

$$KE = 5.00 \times 10^{-13} \text{ J}$$

$$\begin{aligned} \text{(b)} \quad E_0 &= mc^2 \\ &= 9.11 \times 10^{-31} \times (3.00 \times 10^8)^2 \end{aligned}$$

$$E_0 = 8.20 \times 10^{-14} \text{ J}$$

$KE/E_0 = 6.09$ — the electron's kinetic energy is over six times its rest energy.

Classical prediction:

$\frac{1}{2}mv^2 = 4.05 \times 10^{-14} \text{ J}$ — off by more than $10\times$.

► What is the total energy E of this electron?

Example: Rest Energy of 1 gram

Mass–Energy Equivalence

Calculate the rest energy of $m = 1.00 \times 10^{-3}$ kg. How does this compare to chemical energy releases?

$$\begin{aligned} E_0 &= mc^2 \\ &= 1.00 \times 10^{-3} \text{ kg} \times (3.00 \times 10^8 \frac{\text{m}}{\text{s}})^2 \end{aligned}$$

$$E_0 = 9.00 \times 10^{13} \text{ J}$$

$9.00 \times 10^{13} \text{ J} \approx 21$ kilotons of TNT — the yield of the first atomic bomb.

Chemical reactions release $\sim 10^{-10}$ of this rest energy. c^2 is enormous.

► The Sun loses about 4×10^9 kg per second to radiation. How much power does it emit?

Example: Speed from Energy

Total Energy to Speed

A particle has rest energy $E_0 = 1.00$ MeV and total energy $E = 4.00$ MeV. Find its (a) kinetic energy, (b) γ , and (c) speed v/c .

(a) $KE = E - E_0$

$$KE = 3.00 \text{ MeV}$$

(b) $\gamma = \frac{E}{E_0} = \frac{4.00}{1.00}$

$$\gamma = 4.00$$

(c) $v^2/c^2 = 1 - 1/\gamma^2$
 $= 1 - 1/16 = 15/16$

$$v = \frac{\sqrt{15}}{4} c \approx 0.968c$$

When total energy is 4 times rest energy, the particle travels at $0.968c$ — extremely close to c but still below it.

► Would the classical prediction for speed be larger or smaller? Why?

Check Your Understanding

A proton ($m_p c^2 = 938 \text{ MeV}$) has total energy $E = 2000 \text{ MeV}$. What is its kinetic energy and Lorentz factor?

Come to lecture to see the answer.

Photons and the Energy-Momentum Relation

For a photon (massless particle), $m = 0$, so the universal relation becomes:

$$E = pc$$

- ▶ Photons have *zero rest mass* but non-zero energy and momentum
- ▶ Energy: $E = h\nu$ (Planck's relation), where ν is frequency
- ▶ Momentum: $p = E/c = h\nu/c = h/\lambda$ (de Broglie)
- ▶ Photons always travel at exactly c
- ▶ There is no rest frame for a photon

The energy-momentum relation $E^2 = (pc)^2 + (mc^2)^2$ works for all particles — massive and massless.

Photons are the only massive-free particles in Nature. Everything else has $m > 0$ and $v < c$.

The Energy Hierarchy Revisited

The fundamental relation for all particles is:

$$E^2 = (pc)^2 + (mc^2)^2$$

- ▶ This is the **energy-momentum relation**. It always holds.
- ▶ For a particle at rest ($p = 0$):
 $E = mc^2$ (*rest energy*)
- ▶ For a moving particle:
 $E = \sqrt{(pc)^2 + (mc^2)^2} = \gamma mc^2$
- ▶ For a massless particle ($m = 0$):
 $E = pc$ (photons and gravitons)

Why This Matters

This relation is the same in *all* inertial frames. It is a fundamental invariant of spacetime, not dependent on any observer's choice of reference frame.

Einstein's Genius

He unified energy, momentum, mass, and the speed of light into a single geometric relation that emerges from the Lorentz transformation.

Problem-Solving Strategy: Energy-Momentum Method

For any relativistic problem involving particles, energy, or momentum:

The Steps

- ▶ Identify given quantities: m , v (or γ , p , E)
- ▶ Calculate $\gamma = 1/\sqrt{1 - v^2/c^2}$
- ▶ Use $E = \gamma mc^2$ for total energy
- ▶ Use $p = \gamma mu$ for momentum
- ▶ Check: $E^2 = (pc)^2 + (mc^2)^2$
- ▶ Apply conservation laws (before/after collision)

The energy-momentum relation is your universal tool. It works in all reference frames and for all particles.

Always verify dimensions and check limiting cases ($v \rightarrow 0$ or $m \rightarrow 0$).

Spacetime Interval: The True Invariant

While observers disagree on space and time separately, they all agree on the **spacetime interval**:

$$s^2 = -c^2(\Delta t)^2 + (\Delta x)^2$$

Example

Two events separated by $\Delta x = 300$ m and $\Delta t = 1 \mu\text{s}$.

- ▶ Observer A: $s^2 = -(300)^2 + (300)^2 = 0$
- ▶ (Photon-like separation)
- ▶ Observer B (moving): will compute *different* $\Delta x'$ and $\Delta t'$, but *same* s^2

The Deep Principle

This invariance of the spacetime interval is the mathematical heart of special relativity. It is the statement that all observers live in the same spacetime geometry.

When Relativity Matters: A Practical Guide

When should you use relativistic equations instead of classical?

Regime	Use
$v < 0.1c$	Classical is fine
$0.1c < v < 0.5c$	Relativity recommended
$v > 0.5c$	Relativity essential
Particle accelerators	Always relativistic
Nuclear reactions	Always relativistic
GPS satellites	Both SR and GR needed

Error Estimates

At $v = 0.1c$: $\gamma = 1.005$ (0.5% error)

At $v = 0.5c$: $\gamma = 1.155$ (15% error)

At $v = 0.9c$: $\gamma = 2.294$ (130% error)

Classical physics is the $v \ll c$ limit of relativistic physics.

Real-World Applications of Special Relativity

GPS	Particle Physics	Nuclear Energy
▶ Satellite clocks run fast (GR) and slow (SR) ▶ Net error: $+38 \mu\text{s}/\text{day}$ ▶ Must be corrected or position drifts $> 10 \frac{\text{km}}{\text{d}}$	▶ LHC protons at $0.99999991 c$ ▶ $E = 6.5 \text{ TeV}$ per proton ▶ $KE \approx 7000 E_0$	▶ Fission/fusion converts $\sim 0.1\%$ of mass ▶ Powers the Sun for 10 billion years ▶ Powers reactors and weapons

The Common Thread

All applications rely on γ , $E = \gamma mc^2$, and $p = \gamma mu$. Special relativity is not exotic theory — it is *essential engineering*.

Real-World Applications, continued

Relativity in Action

Without relativity, modern technology fails: GPS systems drift $10 \frac{\text{km}}{\text{d}}$, particle accelerators cannot be designed, nuclear power cannot be explained, and we could not understand stars.

Closing the Loop

Remember the opening question: if a spaceship travels at $0.9c$ and turns on a headlight, what do Earth observers measure?

The answer — c , not $1.9c$ — encapsulates the entire chapter.

Thinking Relativistically

Special relativity teaches us to think about physics differently.

- ▶ Observers don't agree on space and time separately
- ▶ But they agree on the laws of physics
- ▶ And they agree on spacetime geometry
- ▶ Invariants are more fundamental than measurements
- ▶ E and p separately depend on frame, but m doesn't
- ▶ Energy and mass are two sides of the same coin
- ▶ c is not just a speed — it is the speed of causality

Special relativity is not about exotic effects. It is about the *deep structure* of nature.

Einstein showed that what seemed like different phenomena (time dilation, length contraction, $E = mc^2$) are all facets of one unified theory.

Chapter Summary: Special Relativity

Topic	Key Equation	Meaning
Lorentz factor	$\gamma = (1 - v^2/c^2)^{-1/2}$	Magnification
Time dilation	$\Delta t = \gamma \Delta t_0$	Moving clocks slow
Length contraction	$L = L_0/\gamma$	Moving rulers shorter
Rel. Doppler	$\lambda_{\text{obs}} = \lambda_s \sqrt{\frac{1+u/c}{1-u/c}}$	Redshift/blueshift
Velocity addition	$u = (v + u')/(1 + vu'/c^2)$	Speed limit c
Rel. momentum	$p = \gamma mu$	Diverges as $v \rightarrow c$
Kinetic energy	$KE = (\gamma - 1)mc^2$	Work to accelerate
Rest energy	$E_0 = mc^2$	Mass $\Leftrightarrow$ energy
Total energy	$E = \gamma mc^2$	Full energy

Unifying Principle

All results flow from two postulates. Space and time are not separate absolutes — they are unified into **spacetime** by the Lorentz transformation.

Chapter Summary: Special Relativity, continued

Looking Forward: General Relativity

Special relativity handles inertial frames. General relativity extends this to *all* reference frames, including accelerating ones. Gravity emerges as curvature of spacetime.

Expert-Level Insight

Special relativity is fundamentally about preserving the same laws of physics for all observers. It is not mainly about clocks and rulers — it is the *geometry* of spacetime itself.

Key Takeaways

As you leave this chapter, remember:

- ▶ The speed of light is the same in all inertial frames
- ▶ This requirement forces time and space to be relative
- ▶ The Lorentz factor γ quantifies all relativistic effects
- ▶ Energy and mass are interconvertible:
 $E = mc^2$
- ▶ Momentum and energy are not absolute; only their combination (the spacetime interval) is invariant
- ▶ Relativity is not exotic — it is the foundation of modern technology

Special relativity overturned 300 years of Newtonian thought. It revealed that space, time, and gravity are more subtle and profound than classical physics suggested.